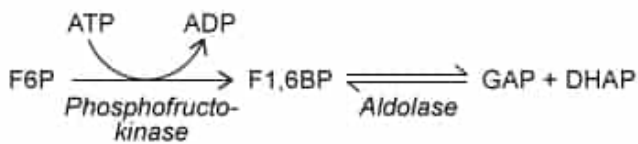


During glycolysis, phosphofructokinase converts fructose 6-phosphate (F6P) into fructose 1,6-bisphosphate (F1,6BP), as shown below. Aldolase then converts F1,6BP into glyceraldehyde 3-phosphate (GAP) and dihydroxyacetone phosphate (DHAP).



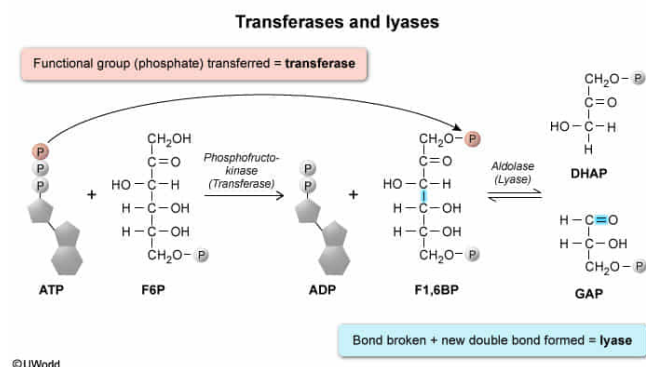
Respectively, phosphofructokinase and aldolase are examples of:

- ☐ A. an isomerase and a hydrolase. [5%]
- ✓ ☒ B. a transferase and a lyase. [84%]
- ☐ C. a ligase and an isomerase. [5%]
- ☐ D. an oxidoreductase and a transferase. [4%]

Correct

83% answered correctly

Explanation:



Enzymes are classified into one of **six categories** based on their mechanism of catalysis. **Transferases** move a chemical group from one molecule to another. Kinases are a type of transferase that either move a phosphate group from ATP to a substrate to form ADP and phosphorylated substrate, or from a substrate to ADP to form ATP. **Phosphofructokinase** transfers a phosphate from ATP to fructose 6-phosphate (F6P), forming ADP and fructose 1,6-bisphosphate (F1,6BP). Therefore, **phosphofructokinase is a transferase**.

Lyases either form two molecules by breaking a bond within one molecule or they combine two molecules into one. To form bonds between reactant molecules, lyases add groups to a double bond. In the reverse reaction, they

break a single bond in a reactant and generate a new double bond in the products. Aldolase splits F1,6BP into glyceraldehyde 3-phosphate (GAP) and dihydroxyacetone phosphate (DHAP) by forming a new C=O double bond. Therefore, **aldolase is a lyase**.

(Choices A and C) **Isomerases** change the positions of functional groups *within* a single molecule, and **hydrolases** break bonds by *adding water* to them. Phosphofructokinase and aldolase transfer functional groups *between* molecules and break bonds by *creating new double* bonds, respectively. **Ligases** use the energy from ATP hydrolysis to combine two molecules into one.

(Choice D) **Oxidoreductases** participate in redox reactions by transferring electrons from one molecule to another, and usually require NAD(P)H or FADH₂ as a substrate or a product. Neither phosphofructokinase nor aldolase catalyze redox reactions.

Educational objective:

Enzymes are classified based on their mechanism of catalysis.

Oxidoreductases move electrons, transferases move functional groups between molecules, hydrolases break bonds by adding water, lyases break bonds by forming a double bond elsewhere, isomerases move functional groups within a molecule, and ligases join molecules together using ATP as an energy source.

Time spent:0 sec

Subject:
Biochemistry

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System:
Enzymes